

token_embedding
Index lookup from weight matrix
 $W_{te} \in \mathbb{R}^{\text{vocab.size} \times d_{model}}$

position_embedding
Index lookup from weight matrix
 $W_{pe} \in \mathbb{R}^{\text{seq.len} \times d_{model}}$

LayerNorm Definitions

Calculated across feature dim d_{model} :

$$\mu = \frac{1}{d_{model}} \sum_{i=1}^{d_{model}} x_i$$

$$\sigma = \sqrt{\frac{1}{d_{model}} \sum_{i=1}^{d_{model}} (x_i - \mu)^2 + \epsilon}$$

blocks.0

blocks.0.ln1 (LayerNorm)
 $X' = \frac{X_0 - \mu}{\sigma} \gamma + \beta$
 $\gamma, \beta \in \mathbb{R}^{d_{model}}$

blocks.0.sa.q_proj
 $Q = X'W_q + b_q$
 $W_q \in \mathbb{R}^{d_{model} \times d_{model}}$
 $b_q \in \mathbb{R}^{d_{model}}$

blocks.0.sa.k_proj
 $K = X'W_k + b_k$
 $W_k \in \mathbb{R}^{d_{model} \times d_{model}}$
 $b_k \in \mathbb{R}^{d_{model}}$

blocks.0.sa.v_proj
 $V = X'W_v + b_v$
 $W_v \in \mathbb{R}^{d_{model} \times d_{model}}$
 $b_v \in \mathbb{R}^{d_{model}}$

F.scaled_dot_product_attention
Computed via FlashAttention kernels: $\text{softmax}\left(\frac{QK^T}{\sqrt{\text{head.dim}}}\right)V$
where $\text{head.dim} = d_{model}/n_{heads}$

blocks.0.sa.out_proj
 $O = \text{Attention} \cdot W_o + b_o$
 $W_o \in \mathbb{R}^{d_{model} \times d_{model}}$
 $b_o \in \mathbb{R}^{d_{model}}$

Residual ($X_{in} = X_0$)

blocks.0.ln2 (LayerNorm)
 $X'' = \frac{X_{res1} - \mu}{\sigma} \gamma + \beta$
 $\gamma, \beta \in \mathbb{R}^{d_{model}}$

blocks.0.ffwd.net.0
 $H_1 = X''W_1 + b_1$
 $W_1 \in \mathbb{R}^{d_{model} \times d_{ff}}$
 $b_1 \in \mathbb{R}^{d_{ff}}$

blocks.0.ffwd.net.1 (ReLU)
 $H_2 = \max(0, H_1)$

blocks.0.ffwd.net.2
 $F = H_2W_2 + b_2$
 $W_2 \in \mathbb{R}^{d_{ff} \times d_{model}}$
 $b_2 \in \mathbb{R}^{d_{model}}$

Residual (X_{res1})

blocks.1 (Identical architecture; X_{b0} serves as input & initial residual)

blocks.2 (Identical architecture; X_{b1} serves as input & initial residual)

blocks.3 (Identical architecture; X_{b2} serves as input & initial residual)

blocks.4 (Identical architecture; X_{b3} serves as input & initial residual)

blocks.5 (Identical architecture; X_{b4} serves as input & initial residual)

blocks.6 (Identical architecture; X_{b5} serves as input & initial residual)

blocks.7 (Identical architecture; X_{b6} serves as input & initial residual)

blocks.8 (Identical architecture; X_{b7} serves as input & initial residual)

blocks.9 (Identical architecture; X_{b8} serves as input & initial residual)

blocks.10 (Identical architecture; X_{b9} serves as input & initial residual)

blocks.11 (Identical architecture; X_{b10} serves as input & initial residual)

ln.f (Final LayerNorm)
 $X_{final} = \frac{X_{b11} - \mu}{\sigma} \gamma + \beta$
 $\gamma, \beta \in \mathbb{R}^{d_{model}}$

output (Linear Projection)
 $\text{Logits} = X_{final}W_{out} + b_{out}$
 $W_{out} \in \mathbb{R}^{d_{model} \times \text{vocab.size}}$
 $b_{out} \in \mathbb{R}^{\text{vocab.size}}$

Color Key

- Linear / Embed / Norm
- Attention Sub-layer
- Feed-Forward Sub-layer
- Stacked Transformer Block